AI FOR CREDIT RISK INTELLIGENCE

BA

BreadAhead

Early Warning System · Project documentation & supporting files

Explainable credit-deterioration monitoring: data validation, rule and ML signals, an Early Warning Score, watchlist, AI-assisted analysis and append-only analyst review.

Repository

github.com/X3PH1RE/BreadAhead

Version

main · sample portfolios as of 31 October 2026

Stack

Python · FastAPI · pandas · scikit-learn · XGBoost · SQLite · React · TypeScript · OpenAI

Contents

1. Executive summary 3

2. How the brief is covered 4

Level 1 required capabilities . . . 4

The eight early-warning scenarios . . . 4

3. Architecture 5

Analyst flow through the product . . . 5

Technology . . . 5

4. Product walkthrough 6

5. Feature reference 11

5.1 Data ingestion and validation . . . 11

5.2 Early warning signals . . . 11

5.3 Early Warning Score . . . 11

5.4 Region profiles . . . 12

5.5 Unusual-behaviour detection (machine learning) . . . 13

5.6 Predicted serious delinquency (machine learning) . . . 13

5.7 Portfolio concentration and clusters . . . 13

5.8 Recommended actions . . . 13

5.9 AI assistants . . . 14

5.10 Watchlist, review and audit . . . 14

6. Validation and results 15

Synthetic data . . . 15

7. Security, privacy and governance 16

8. Running the software 16

9. Limitations and next steps 17

Appendix A · API 17

Appendix B · Repository layout 17

1. Executive summary

Lenders usually see credit trouble only once a payment is missed. **BreadAhead**, our Early Warning System (EWS), watches how each customer's behaviour changes after the loan is made: repayments, cash flow, borrowing and, for small businesses, business health. It flags deterioration early, explains every point of its score, recommends who should act and by when, and gives analysts a defensible, append-only way to record decisions.

Capability	What the software does
Ingestion & validation	CSV, Excel or JSON; every record checked and linked; bad rows quarantined, derivable values repaired, the rest flagged. AI or fuzzy mapping for a bank's own column names.
Early warning signals	15 configurable rule signals over time windows, plus an isolation-forest model that flags unusual behaviour no rule anticipated.
Warning score	0–100, higher = riskier. Four risk areas combined, with adjustments for one-off misses, seasonality, holidays and roll-forward risk; confidence and an honest 'Insufficient data' status.
Prediction	XGBoost estimate of reaching 30+ DPD within three months, shown beside (never changing) the rule score, with per-factor explanations.
Watchlist & portfolio	Filterable, sortable watchlist with next actions; statistically tested risk clusters; Ask EWS turns plain-English questions into filters.
Interventions	Rules playbook recommends actions with owner and due date; AI case notes, cluster briefings and a portfolio risk memo explain and prioritise.
Analyst review	The four decisions in the brief, reasons required, full history; original score and evidence can never be edited. AI evidence check for documents.

Design principle. Every number is traceable. Rules produce the score; machine learning adds context beside it; AI drafts explanations that are checked in code against the evidence. Nothing automated can change a saved score or overrule the analyst.

2. How the brief is covered

Level 1 required capabilities

Requirement (brief)	Where it lives
1. Customer & credit data ingestion: upload or select; CSV, Excel, JSON; validate; link customers, facilities, repayments and indicators	Data & validation page; **ews/engine/loader.py, validate.py**; sample portfolios; column mapping
2. Early warning signal detection, using configurable rules, statistics or ML	15 rule signals (config/risk_signal_definitions.json), UB01 isolation forest, XGBoost forecast
3. Warning score with explanation of factors	0–100 score, area breakdown, per-signal points that add up to the score, adjustments in plain English
4. Watchlist with filters: product, segment, risk level, delinquency status, portfolio	Watchlist page; also region, industry, customer type, active signal, score range, review status; CSV export
5. Analyst review: Continue Monitoring, Requires Investigation, False Positive, Escalate for Review, with reason; preserve original score and evidence	Customer page review panel; SQLite triggers reject UPDATE/DELETE; original alert downloadable as JSON

The eight early-warning scenarios

Scenario	How EWS handles it
1. Repayment deterioration	Late frequency, DPD trend slope, partial payments, current DPD; isolated delay vs recurring pattern
2. Declining income & cash flow	Consecutive income declines, 3-month vs prior averages, balance erosion, outflows above inflows, bounced payments
3. Rising obligations & exposure	Utilisation rise, new facilities, obligations growing faster than inflows, credit inquiries
4. Temporary misses & exceptions	One-off dampener (×0.25 with evidence, ×0.6 without), holiday calendar exceptions, AI evidence check
5. Early delinquency & roll-forward	Roll-forward rule (score ≥ 75), recent-episode rule, recovering vs worsening distinguished; ML 30+ DPD forecast
6. SME & business stress	SME signals: late tax filings, receivable days rising; cash-flow and exposure signals
7. Missing or incomplete history	Signals marked 'missing data', confidence share, 'Insufficient data' instead of a risk band; missing data never adds points
8. Portfolio clusters	Binomial tests by industry×region, industry, region, segment, portfolio and product; shared-signal clusters; AI briefings

3. Architecture

One command runs everything: **python -m api** starts FastAPI, which serves both the JSON API and the built React front end. All scoring, ML inference and storage run locally; only the optional AI features call OpenAI.

Layer	Components
Web app (web/)	React 18 + TypeScript (Vite), React Query, Recharts. Pages: Overview, Watchlist, Customer, Portfolio & clusters, Risk memo, Data & validation, Audit trail, Settings, How scoring works.
API (api/)	FastAPI. Dataset registry (content-hashed), presenters, AI modules (case notes, Ask EWS, briefings, memo, mapping, evidence), serves the built web app.
Engine (ews/engine/)	Loader, validator, customer-month views, 16 signals, scoring, explanations, clusters, unusual-behaviour model. Configured by config/*.json.
ML (ews/ml/)	Point-in-time feature reconstruction, XGBoost training CLI, prediction service with TreeSHAP factors; versioned models in models/.
Evidence store (ews/store/)	SQLite: runs, alerts, reviews, AI notes and outputs, evidence decisions, ML predictions; append-only by database triggers.
Data (data/, scripts/)	Synthetic India (500) and GCC Islamic (400) portfolios with clean, dirty and ground-truth files; generator script.

Analyst flow through the product

Data & validation → Assessment (score) → Overview → Watchlist → Customer → Review

- **Data & validation:** load a sample or upload files; review the validation report; run an assessment (about 20–25 s for 500 customers).
- **Overview:** exposure at risk, bands, threshold curve, review progress, open actions, clusters.
- **Watchlist:** who needs attention; Ask EWS; filters; next action per customer; CSV export.
- **Customer:** why the score is what it is, ML forecast, case note, recommended actions, evidence charts, repayment history.
- **Review:** one of the four decisions with a reason; history kept; evidence decisions recorded.
- **Portfolio, Risk memo, Audit trail:** clusters and briefings, the management memo, every saved assessment and decision.

Technology

Area	Choices
Back end	Python 3.12, FastAPI, Uvicorn, pandas, NumPy, openpyxl, SQLite
Machine learning	scikit-learn (IsolationForest), XGBoost (gradient-boosted trees, native TreeSHAP)
AI	OpenAI chat completions with structured outputs (Pydantic schemas); default model gpt-4.1-mini, configurable
Front end	React 18, TypeScript, Vite 6, React Router 7, TanStack Query, Recharts, Lucide icons, Outfit font
Quality	pytest (98 tests), TypeScript strict type-check, npm audit clean at time of writing

4. Product walkthrough

Screens from the running application on the India sample portfolio (500 customers, as of 31 Oct 2026).

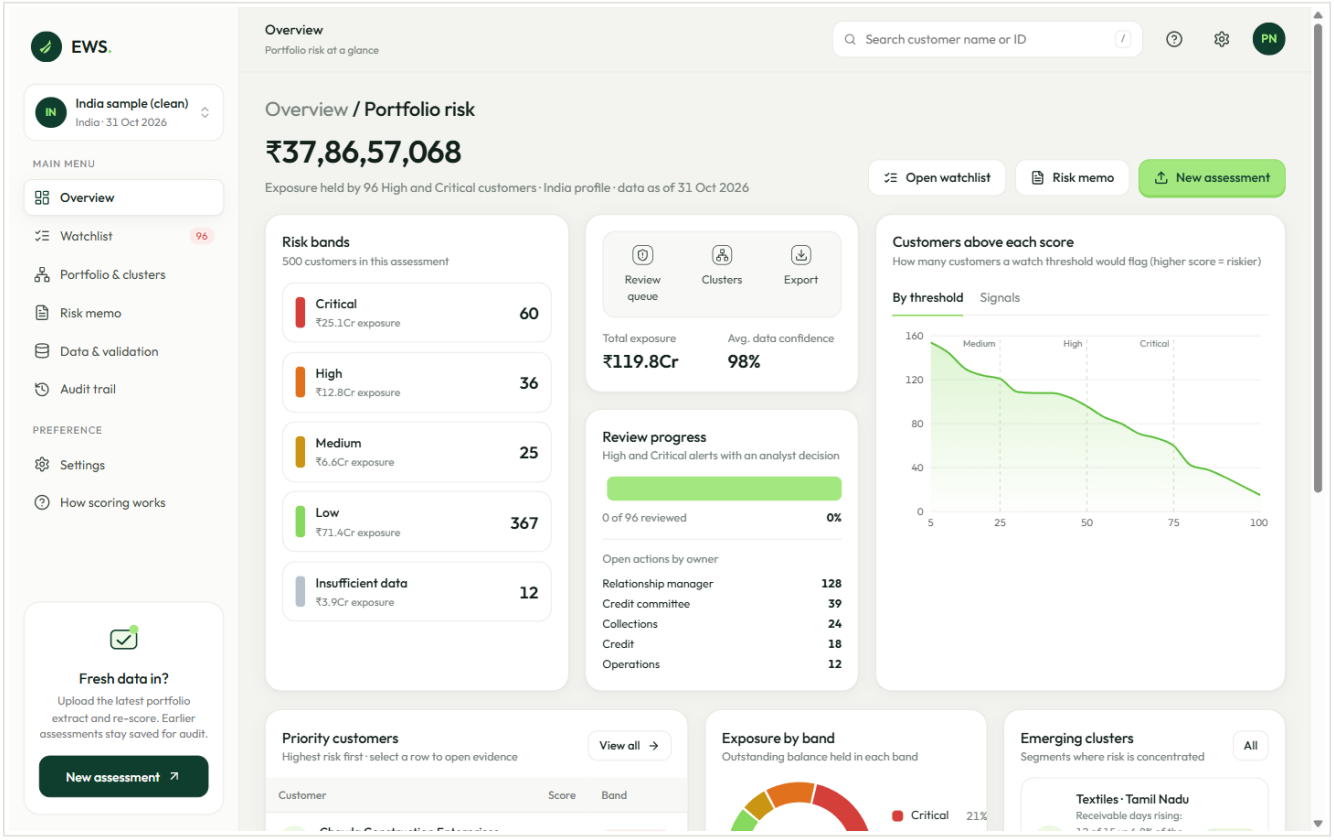


Figure 1 · Overview: exposure held by High and Critical customers, risk bands, customers above each score threshold, review progress and quick actions.

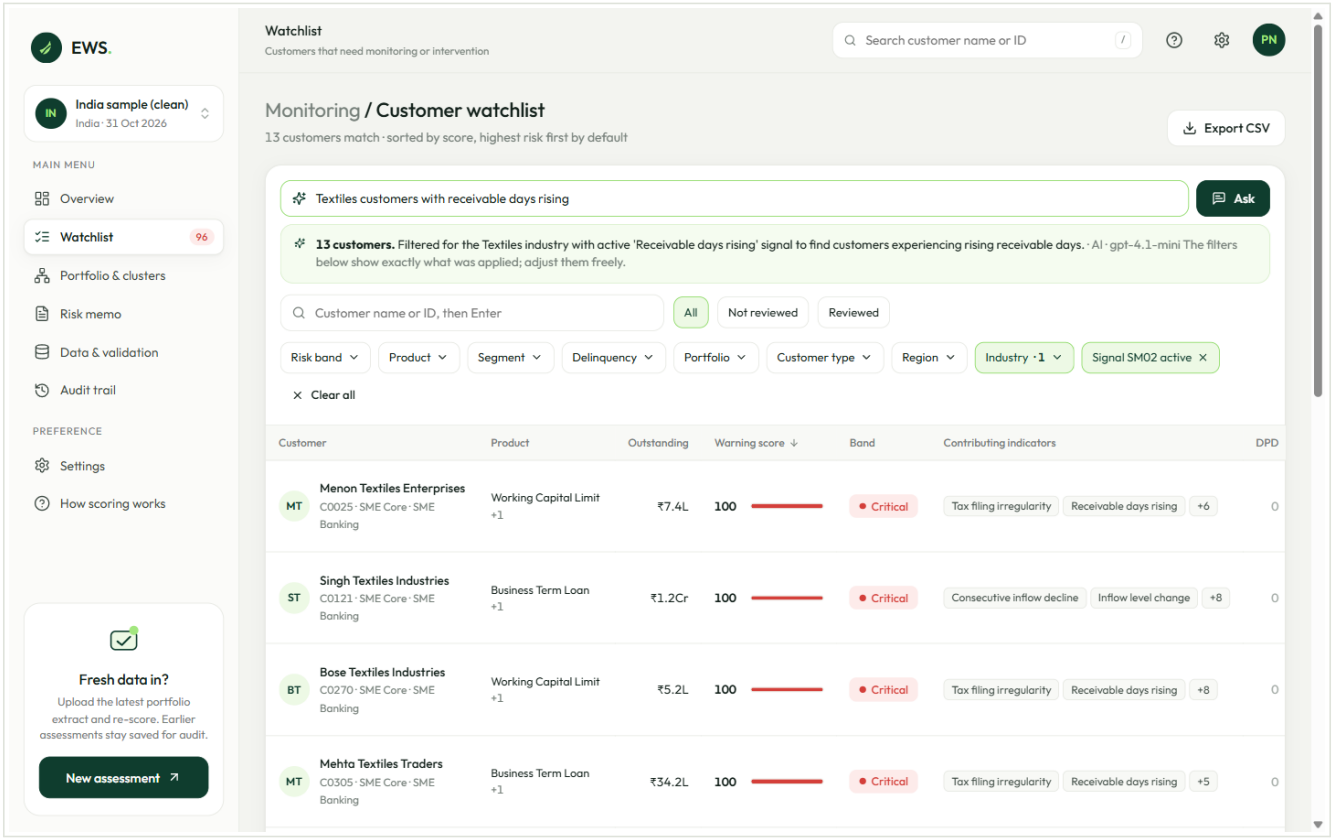


Figure 2 · Watchlist with Ask EWS: "Textiles customers with receivable days rising" became the filters Industry = Textiles and active signal SM02; the applied filters stay visible and editable.

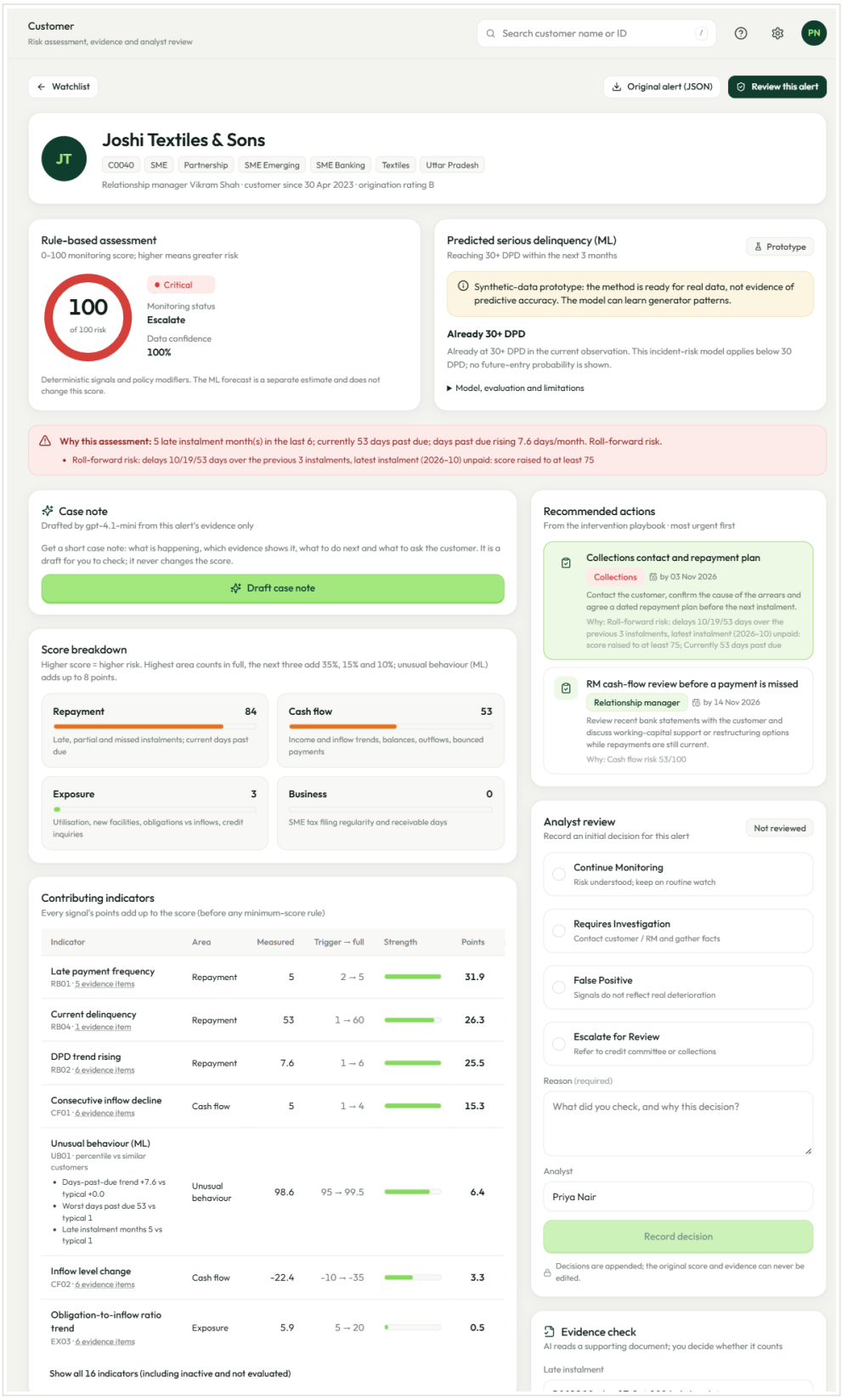


Figure 3 · Customer page (Joshi Textiles & Sons, Critical 100): rule score and ML forecast side by side, roll-forward explanation, case note, recommended actions (collections within 3 days), analyst review and every contributing indicator including the unusual-behaviour signal.

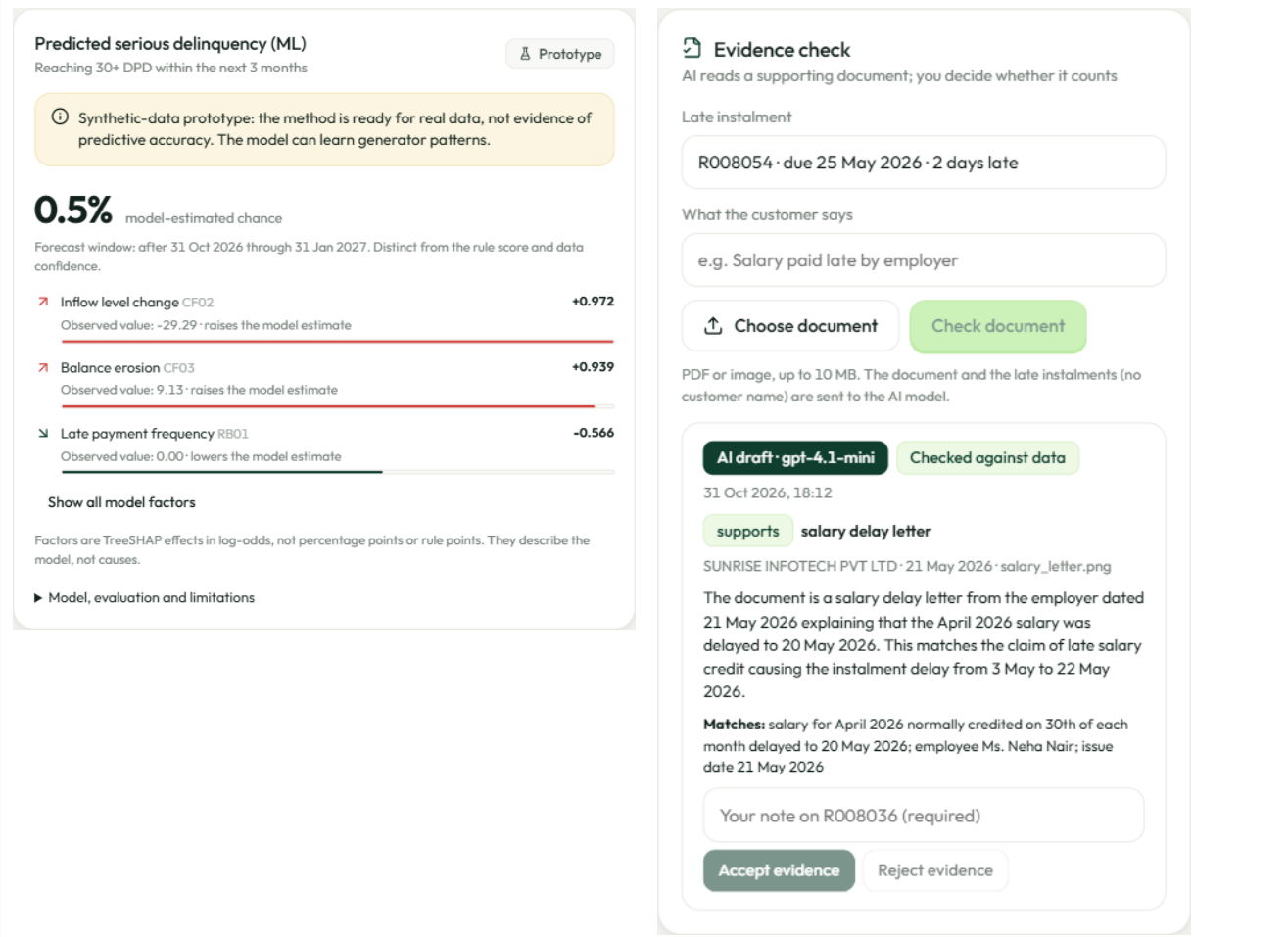


Figure 4 · Left: ML forecast card (probability of reaching 30+ DPD within three months, with factors and caveats). Right: AI evidence check: a salary-delay letter judged "supports" and matched to the late instalment; the analyst accepts or rejects it.

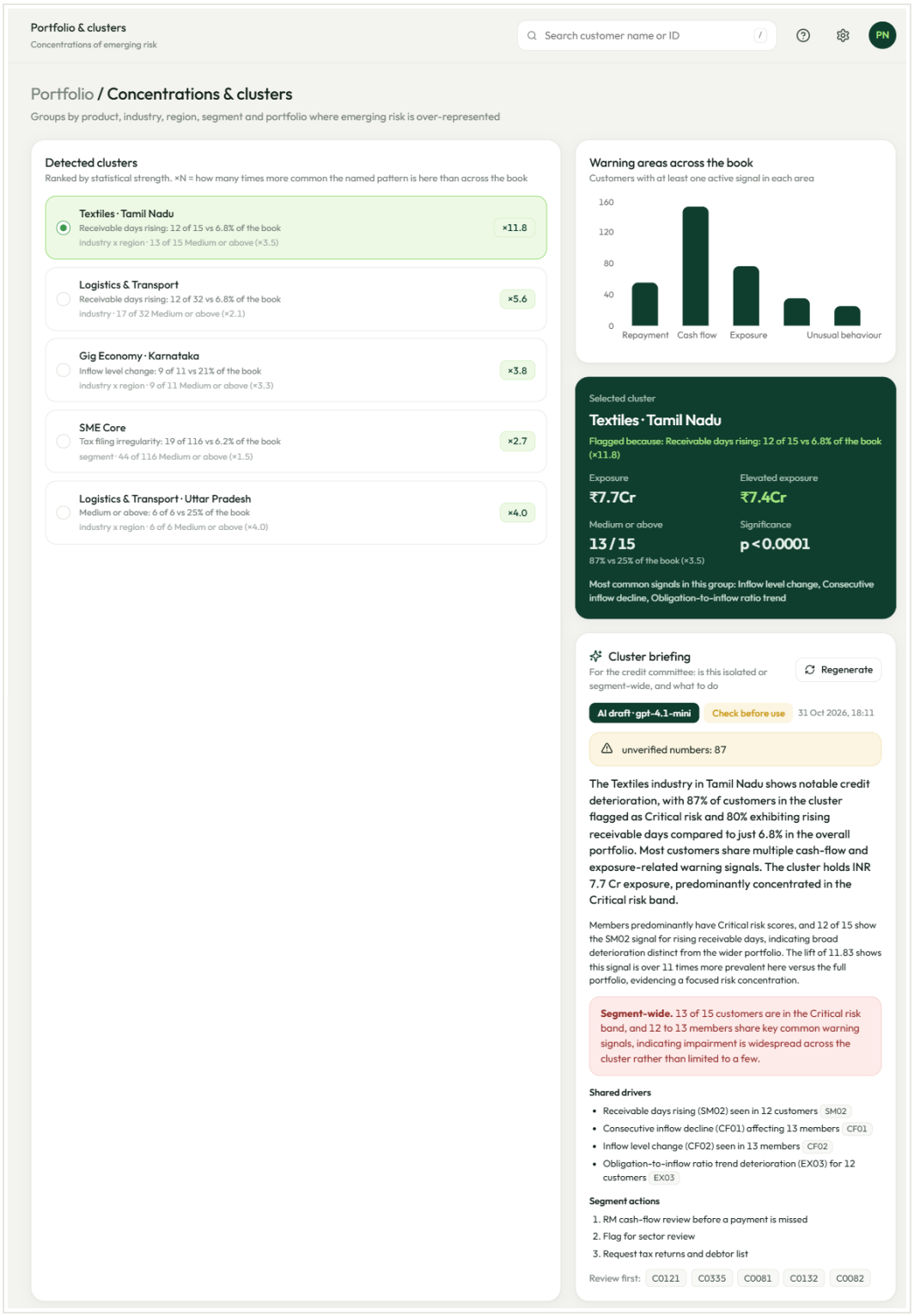


Figure 5 · Portfolio & clusters: detected concentrations, what each multiplier measures, warning areas across the book, and the AI cluster briefing (segment-wide verdict, shared drivers, segment actions, customers to review first).

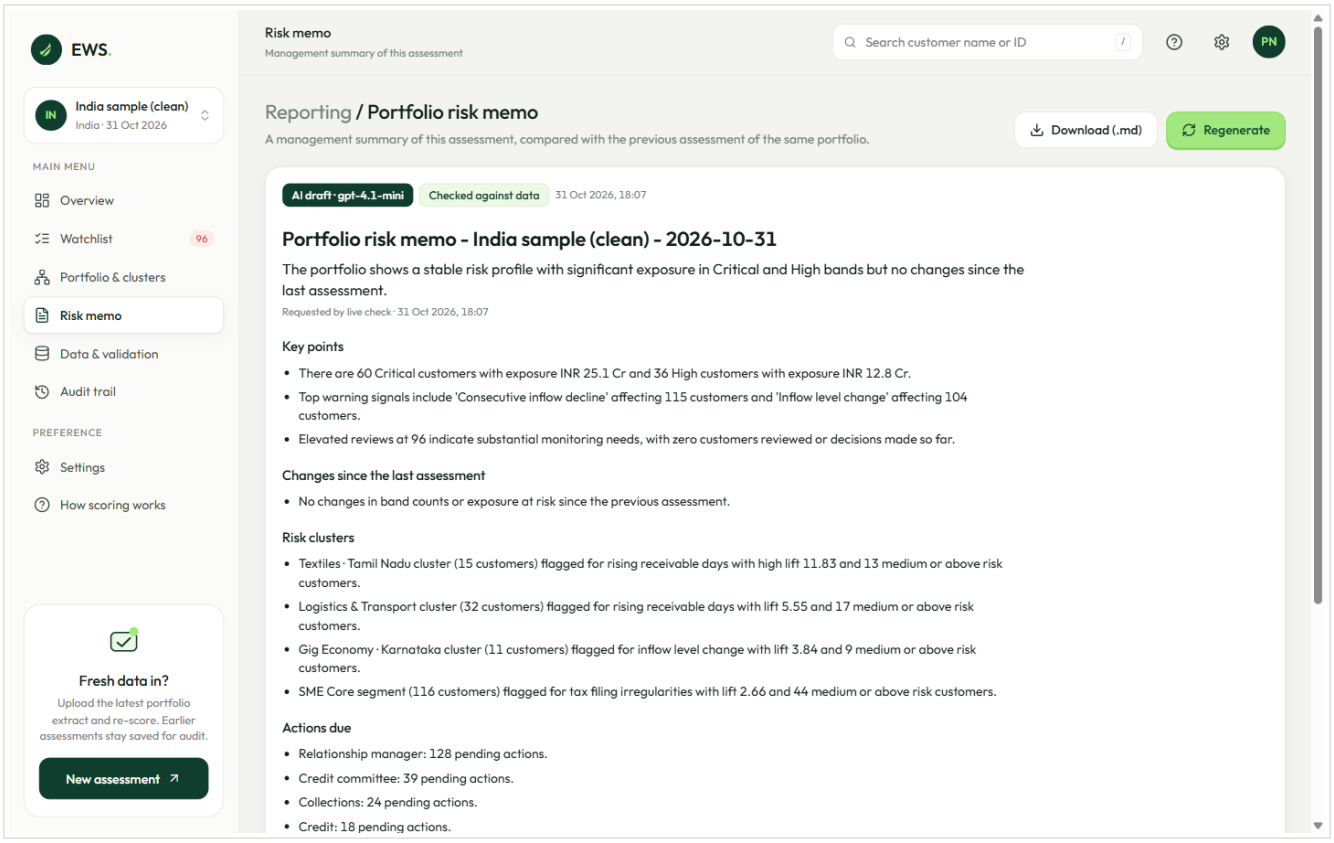


Figure 6 · Risk memo: AI-drafted management summary, checked against the assessment's figures and compared with the previous assessment; downloadable as Markdown.

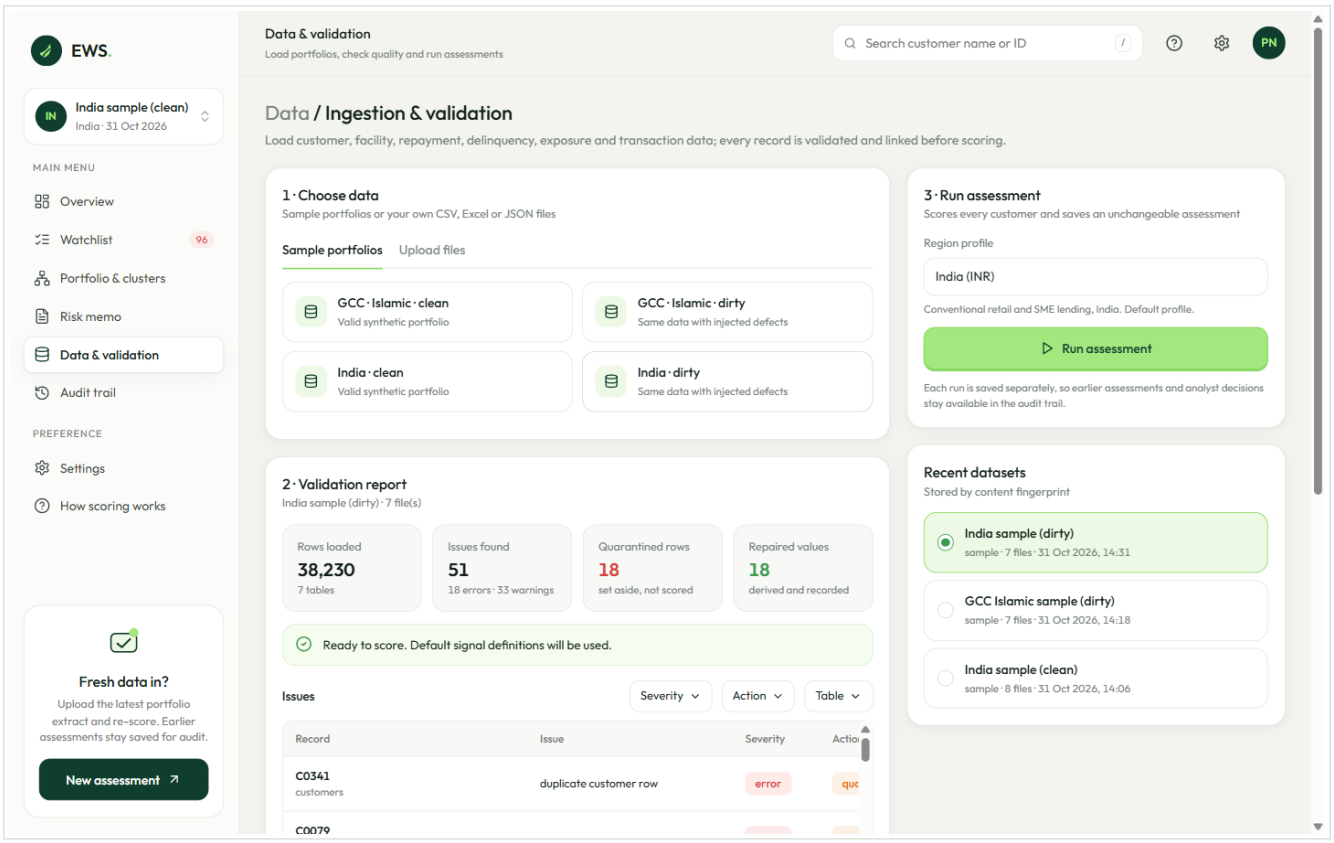


Figure 7 · Data & validation on the dirty sample: 51 issues found, 18 rows quarantined, 18 values repaired, ready to score.

5. Feature reference

5.1 Data ingestion and validation

- **Inputs:** one CSV per table, one Excel workbook (one sheet per table) or one JSON file. Tables: customers, facilities, repayments, delinquency, exposure, transactions, optional sme_indicators, optional risk_signal_definitions (overrides thresholds and weights for that upload).
- **Checks:** schema and types, allowed values, ranges, duplicates, links between tables (orphan loans, repayments, transactions), contradictory values (e.g. 'Paid On Time' with DPD > 0).
- **Outcomes:** quarantined (unusable rows set aside with the reason), repaired (e.g. total inflows recalculated, utilisation recomputed, '1,20,000' parsed, customer type fixed) or kept with the bad field blanked. Every outcome is listed in the validation report; quarantined rows can be inspected.
- **Datasets** are stored by content hash so every saved assessment can be traced to its exact files.
- **Column mapping:** for a bank's own file and column names, AI (or fuzzy matching without a key) proposes the table and column for each file; the analyst edits and approves it before upload.

5.2 Early warning signals

Each signal's strength rises from 0 at its trigger to 1 at full strength; points = weight × strength. Signals without enough data are 'missing data', those that don't fit the customer 'not applicable'; neither adds points.

ID	Signal	Fires when	Trigger → full	Weight
RB01	Late payment frequency	Late payments in 2+ of the last 6 months	2 → 5	15
RB02	DPD trend rising	Days past due rising per month	1 → 6 d/mo	12
RB03	Partial payments	Partially paid instalments in 6 months	1 → 3	6
RB04	Current delinquency	An instalment is overdue now	1 → 60 DPD	14
CF01	Falling income	Consecutive monthly declines totalling ≥ 10%	1 → 4 months	14
CF02	Inflow level drop	Last 3 months vs prior 3	−10% → −35%	6
CF03	Balance erosion	Average balance, last 3 vs prior 3	−15% → −50%	5
CF04	Spending more than earning	Outflows above inflows	2 → 5 months	3
CF05	Bounced transactions	Returned payments in 6 months	1 → 4	4
EX01	Utilisation rise	Revolving utilisation over 6 months	+15 → +40 pts	8
EX02	New facilities	New loans or lines (not a new customer's first)	1 → 3	4
EX03	Debt outgrowing income	Obligations ÷ inflows rising	+5 → +20 pts	7
EX04	Credit inquiries	New applications in 90 days	2 → 6	2
SM01	Tax filing problems (SME)	GST/VAT filed late or not filed	1 → 4	6
SM02	Customers paying slower (SME)	Receivable days rising	+10 → +35 days	6
UB01	Unusual behaviour (ML)	Among the most unusual 5% of comparable customers	95 → 99.5 pctl	+8 max

5.3 Early Warning Score

- **Areas:** Repayment, Cash flow, Exposure and Business each score 0–100 from their evaluated signals (an area needs at least half its weight evaluated).
- **Combination:** highest area + 0.35 × second + 0.15 × third + 0.10 × fourth, capped at 100; UB01 then adds up to 8 points. Signal points add up to the score, so every score is traceable.

- **Adjustments** (always listed on the customer page): one-off missed payment ×0.25 with evidence or ×0.6 without; seasonal dip that matches last year ×0.3; weekend/holiday delays up to 3 days treated as exceptions; roll-forward risk raises the score to at least 75; a recent delinquency episode to at least 25.
- **Bands:** Low 0–24 (Routine), Medium 25–49 (Watch), High 50–74 (Investigate), Critical 75–100 (Escalate).
- **Missing data:** confidence = share of applicable signal weight evaluated. Under 4 repayment months or confidence below 50% gives 'Insufficient data' (collect data), never a risk band.

5.4 Region profiles

Region-specific settings live in config/profiles/*.json, not code: India (INR, Sunday weekend, Indian holidays) and GCC Islamic (AED, Saturday–Sunday weekend, Eid holidays, Islamic contract types such as Murabaha, Ijarah and Tawarruq, Shariah-compliant rescheduling action). Thresholds, combination factors and rule settings are configurable per profile.

5.5 Unusual-behaviour detection (machine learning)

An isolation forest (scikit-learn), fitted separately for individuals and SMEs, learns typical behaviour from 14 measures of each customer's last six months: income trend, volatility and change vs the same months last year, balance change, outflows ÷ inflows, bounced payments, utilisation and obligation changes, inquiries, late months, worst DPD, DPD trend, and for SMEs receivable days and late filings. Each customer gets a percentile of unusualness; UB01 triggers at 95 and adds at most 8 points. Flags list the measures furthest from typical (e.g. “Utilisation change +61.6 vs typical +1.1”). Missing values take the group median so missing data never looks unusual; a documented one-off miss reduces the signal; results are repeatable (fixed seed).

5.6 Predicted serious delinquency (machine learning)

An XGBoost model estimates the chance of a new recorded 30+ DPD event within the next three calendar months, using the same 15 rule-signal measurements reconstructed at each historical month-end (point-in-time, no future data). Shown beside the rule score; it never changes the score, band, alert or reviews.

- **Explanations:** native TreeSHAP contributions, calibrated to log-odds, list the factors that raised or lowered the estimate.
- **Guardrails:** customers already at 30+ DPD show ‘Already 30+ DPD’ (no fabricated 100%); insufficient data suppresses a forecast; mid-month dates are refused; models load from versioned, checksummed artefacts; forecasts are stored append-only.
- **Shipped model (synthetic data):** customer-group split; held-out ROC AUC 0.84 on only 2 positive rows out of 1,046; Brier score equal to the constant baseline (0.0019). This demonstrates the method, ready for validation on real history; it is **not** a claim of predictive accuracy.

5.7 Portfolio concentration and clusters

Customers are grouped by industry × region, industry, region, segment, portfolio and main product. A group is flagged when its share of Medium-or-above customers (lift ≥ 1.5, p ≤ 0.001) or of customers sharing one warning signal (lift ≥ 2.5, p ≤ 0.0001) is significantly above the book. Broad groups that only echo a stronger cluster inside them are dropped. Each card names what its multiplier measures, e.g. “Receivable days rising: 12 of 15 vs 6.8% of the book (×11.8)”.

5.8 Recommended actions

Action	Owner	Due	Triggered by
Collections contact and repayment plan	Collections	3 days	Roll-forward risk or 30+ DPD
RM call on repayment pattern	Relationship manager	7 days	Repayment area ≥ 30
Reschedule without additional profit	Credit	7 days	GCC Islamic profile with repayment stress
Freeze limit increases, review debt burden	Credit	7 days	Exposure area ≥ 50
RM cash-flow review before a missed payment	Relationship manager	14 days	Cash-flow area ≥ 50
Request tax returns and debtor list	Relationship manager	14 days	SM01 or SM02 active
Ask for evidence of the missed payment	Relationship manager	14 days	Undocumented one-off delay of 7+ days
Flag for sector review	Credit committee	30 days	Member of an industry, region or product cluster
Sense-check unusual account behaviour	Relationship manager	30 days	Only UB01 fired
Collect missing data	Operations	30 days	Insufficient data

Each recommendation lists the reasons and signal IDs that triggered it. The watchlist shows each customer's next action and the Overview counts open actions by owner.

5.9 AI assistants

All AI features share one design: limited input (no customer names unless EWS_AI_SEND_NAMES=1), a fixed output schema via OpenAI structured outputs, checks in code after drafting, add-only storage with reuse for identical input, and a fallback that works without an API key.

Feature	Where	What it does	Checks in code	Without a key
Case note	Customer	Summary, cited findings, ranked next steps, questions for the customer, data gaps, suggested decision; 'Use in review' prefills the form	Citations must exist; actions must be in the playbook; numbers must be in the alert; forbidden wording	Rules draft
Ask EWS	Watchlist	Plain-English question → watchlist filters	Only real filter values kept; 'all values' dropped	Keyword matching
Cluster briefing	Portfolio	Pattern, shared drivers, isolated vs segment-wide, segment actions, customers to review first	Signals and customers must exist; numbers; wording	From cluster figures
Risk memo	Risk memo	Management summary with changes since the last assessment; Markdown download	Numbers; wording	From summary figures
Column mapping	Data	Maps a bank's files and columns to the EWS tables	Valid tables/targets only; no duplicates; required columns	Fuzzy matching
Evidence check	Customer	Reads a PDF or image explaining a late payment; verdict, matches, concerns	Instalments must belong to the customer	Stored for manual review

Every AI output is also scanned for wording the EWS must not use, such as 'default rate' or 'probability of default': the system measures deterioration, not default. Outputs that fail any check are shown with a 'Check before use' warning; nothing AI-generated changes a score or records a decision.

5.10 Watchlist, review and audit

- **Watchlist:** customer, product, outstanding, score, band, contributing indicators, next action, DPD, confidence, monitoring status and review; filters by band, product, segment, delinquency, portfolio, customer type, region, industry, active signal, score range and review status; sorting, paging, shareable URLs, CSV export.
- **Analyst review:** Continue Monitoring, Requires Investigation, False Positive, Escalate for Review; a reason and analyst name are required; each change is a new record; the review notes which case note it used.
- **Evidence decisions:** accept or reject a document for a late instalment, with a note; recorded for audit, the score is unchanged.
- **Preservation:** the original alert (score and all evidence) is stored once and downloadable as JSON; SQLite triggers reject UPDATE and DELETE on runs, alerts, reviews, AI outputs, evidence decisions and forecasts.
- **Audit trail:** every saved assessment with its dataset and region, and every decision in order. Earlier assessments remain selectable from the sidebar switcher.

6. Validation and results

The synthetic generator plants a known scenario for every customer. The answer key (ground_truth.csv) is used only by tests, never as an input. Bands per scenario on the clean portfolios:

Scenario	India L / M / H / C / Insuff.	GCC L / M / H / C / Insuff.
Healthy	307 / 2 / 0 / 0 / 0	252 / 1 / 0 / 0 / 0
Volatile but stable	15 / 0 / 0 / 0 / 0	12 / 0 / 0 / 0 / 0
Seasonal SME	10 / 0 / 0 / 0 / 0	8 / 0 / 0 / 0 / 0
Temporary miss (with / without evidence)	20 / 0 / 0 / 0 / 0	16 / 0 / 0 / 0 / 0
No transaction data · transaction gap	15 / 0 / 0 / 0 / 0	12 / 0 / 0 / 0 / 0
Thin history	0 / 0 / 0 / 0 / 12	0 / 0 / 0 / 0 / 10
Repeated delays	0 / 5 / 2 / 13 / 0	0 / 6 / 4 / 6 / 0
Declining cash flow	0 / 4 / 14 / 0 / 0	0 / 6 / 6 / 2 / 0
Increasing exposure	0 / 2 / 12 / 4 / 0	0 / 3 / 9 / 2 / 0
Early delinquency, recovering	0 / 10 / 0 / 0 / 0	0 / 8 / 0 / 0 / 0
Early delinquency, worsening	0 / 0 / 0 / 12 / 0	0 / 0 / 0 / 10 / 0
SME financial stress	0 / 0 / 0 / 12 / 0	0 / 0 / 0 / 10 / 0

- **No false alarms on safe customers:** 0 of 389 (India) and 0 of 317 (GCC) safe customers rated High or Critical.
- **Clusters:** all five planted clusters found (Textiles·Tamil Nadu, Gig workers·Karnataka, Logistics; Contractors·Dubai, Delivery riders·Riyadh).
- **Validation:** every defect injected in the dirty data is reported; CSV, Excel and JSON produce identical tables.
- **Unusual behaviour:** no healthy customer in the top 5%; median percentile 35 for healthy vs 77–98 for risky scenarios.
- **Automated tests:** 98 pytest tests (engine, validation, scenarios, ground truth, contract, API, interventions, case notes, AI features with fake providers, ML features and model, environment loading) plus a strict TypeScript type-check and production build.
- **Live AI checks** with gpt-4.1-mini: case notes and briefings cite real signals; Ask EWS produced correct filters; a test salary-delay letter was judged ‘supports’ for the right instalment; renamed bank columns were all mapped.

Synthetic data

scripts/generate_data.py writes 18 months of history per portfolio with a realistic mix (about 60% healthy) and every brief scenario plus traps: seasonal businesses, volatile but stable income, one-off misses with and without evidence, missing transaction feeds, roll-forward into 31–60 and 61–90 DPD, and industry/region clusters. Each portfolio comes as clean CSV, Excel and JSON, a dirty copy with a defect manifest (duplicates, orphans, invalid months, separators, wrong case, negative amounts) and the ground-truth file.

7. Security, privacy and governance

Topic	Approach
Immutability	Scores, evidence, reviews, AI outputs, evidence decisions and forecasts are append-only; database triggers reject edits and deletes.
Explainability	Signal points add up to the score; adjustments listed in plain English; ML factors reconcile to the forecast; AI outputs cite signals and evidence.
Human in the loop	Analysts make every decision; AI only drafts, and evidence checks never change a score.
Data sent to OpenAI	Ask EWS: question and filter values. Briefings and memos: aggregated figures and customer IDs. Case notes: the alert's signals and actions. Mapping: headers and 3 sample values per column. Evidence: the document and late instalments. No customer names unless EWS_AI_SEND_NAMES=1.
Secrets	OPENAI_API_KEY in a git-ignored .env file (template .env.example); environment variables take priority.
Answer key	ground_truth.csv is never ingested by the app; samples skip it explicitly.
Model artefacts	ML models in versioned XGBoost UBJ format with checksums; incompatible or invalid artefacts disable the forecast, not the app.

8. Running the software

Requires Python 3.11+ and Node 20+.

Step	Command
Install back end	python -m venv .venv · .venv\Scripts\activate · pip install -r requirements.txt
Build front end	cd web · npm ci · npm run build
Optional AI	copy .env.example to .env and set OPENAI_API_KEY (and EWS_AI_MODEL)
Run	python -m api → http://127.0.0.1:8000 (API docs at /docs)
Develop	python -m api --reload and, in web/, npm run dev (port 5173)
Test	python -m pytest · cd web && npm run typecheck
Retrain forecast	python -m ews.ml.train
Regenerate data	python scripts/generate_data.py --profile india --seed 42

Setting	Default	Purpose
OPENAI_API_KEY	not set	Enables AI features; without it they fall back to rules
EWS_AI_MODEL	gpt-4.1-mini	OpenAI model for all AI features
EWS_AI_ENABLED	1	Set 0 to turn AI off
EWS_AI_SEND_NAMES	not set	Set 1 to include customer names in AI input
EWS_VAR_DIR / EWS_DB_PATH	var/ · var/ews.sqlite3	Stored datasets, evidence documents and the evidence database
EWS_DATA_ROOT	data/	Sample portfolios
EWS_ML_MODEL_DIR	models/delinquency_3m	Forecast model directory

9. Limitations and next steps

- **Synthetic data:** all results come from generated portfolios; thresholds and both ML models need calibration and validation on real bank history.
- **Forecast model:** very few positive outcomes in the synthetic history; the customer-group split is not an out-of-time backtest.
- **Authentication:** analysts enter their name in Settings; production needs single sign-on and role-based access.
- **Integrations:** connections to core banking, credit bureau and account-aggregator feeds, and scheduled re-scoring, are designed for but not built.
- **Front end:** designed for laptop and desktop widths; phone and tablet layouts were not verified.
- **Holiday calendars** in the region profiles are illustrative and should be replaced with official calendars.

Appendix A · API

Interactive documentation is served at /docs when the app is running.

Method	Path	Purpose
GET	/api/config, /api/health	AI status, profiles, decisions, samples
GET/POST	/api/datasets, /api/datasets/sample, /api/datasets/upload	List, load sample, upload (with optional column mapping)
POST	/api/datasets/propose-mapping	Propose table/column mapping for uploaded files
GET	/api/datasets/{id}, …/quarantined/{table}, …/tables/{table}	Validation report, quarantined rows, table preview
GET/POST	/api/runs	List or create assessments
GET	/api/runs/{run}/summary, …/watchlist, …/watchlist.csv, …/clusters	Overview, watchlist (all filters), export, clusters
POST	/api/runs/{run}/ask	Ask EWS
POST/GET	/api/runs/{run}/clusters/briefing(s)	Cluster briefings
POST/GET	/api/runs/{run}/memo, …/memo/{id}.md	Risk memo and Markdown download
GET	/api/runs/{run}/customers/{cid}, …/alert.json	Customer evidence and original alert
POST	/api/runs/{run}/customers/{cid}/reviews	Record an analyst decision
GET/POST	…/customers/{cid}/case-notes	Case notes
GET/POST	…/customers/{cid}/evidence-checks, …/evidence-decisions	Evidence checks and decisions
GET	/api/ml/model, …/customers/{cid}/ml, …/ml/history	Forecast model info, forecast, saved forecasts
GET	/api/runs/{run}/reviews	Decision log

Appendix B · Repository layout

Path	Contents
api/	FastAPI app, dataset registry, presenters, AI modules (ai, case_notes, ask, briefings, mapping, evidence)
ews/engine/	loader, validate, views, signals, scoring, explain, clusters, anomaly, config
ews/ml/	features, train, service (delinquency forecast)
ews/store/db.py	Append-only SQLite evidence store
ews/interventions.py · config/interventions.json	Recommended-action playbook

Path	Contents
config/	Signal definitions and region profiles (india, gcc_islamic)
web/	React + TypeScript front end
models/delinquency_3m/	Versioned forecast model and metadata
data/	India and GCC sample portfolios (clean, dirty, ground truth)
scripts/generate_data.py	Synthetic portfolio generator
tests/	98 automated tests
docs/DEMO_GUIDE.md · ENGINE.md · ML_MODEL.md · README.md	Demo script, engine, model and setup documentation